AI Predicts, ArcGIS Pro Checks: Testing Planning Forecasts Against Reality

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Title	AI Predicts, ArcGIS Pro Checks: Testing Planning Forecasts Against Reality (74/80)
Abstract	AI can forecast redevelopment in seconds, and its maps look convincing. Testing them against reality is hard: records sit on the street, not the lot, and buildings finish years after they start. ArcGIS Pro checks a retrospective AI forecast for 42,000 Manhattan lots, made from 2020 data. Matching 2021-2025 certificates of occupancy to tax lots found 430 candidate completed lots; a plain overlay found 16. AI and a planning rule ranked permits similarly; on completions, a strict timing filter cut the AI's lead from large to uncertain. Planners and GIS analysts get the workflow and a checklist. 598/600 characters. Published as written in the agenda.
Keywords	AI validation; ArcGIS Pro; urban planning; redevelopment forecasting; public records
Topic category	State and local government: Planning
Audience and level	Planners and GIS analysts. Intermediate; no AI background needed.
Primary speaker	Parya Payami, PhD Dissertator, Department of Architecture, University of Wisconsin-Milwaukee
Links	Code and results: github.com/ParyaPayami/proveit Land-use impact tool: planarena-manhattan.onrender.com

Expanded abstract

The problem. AI models now produce redevelopment forecasts in seconds, and the maps look convincing. Testing them against what really happened is where errors hide: public records are placed on the street rather than the lot, buildings are certified years after they start, and each record measures something slightly different from the forecast. A careless test can make a model look better or worse than it is. **The forecasting model is the case study; the ArcGIS Pro workflow is the lesson.**

The case. PlanArena, a parcel-level planning model built as a result of six years of PhD research at the intersection of architecture/urban planning, AI and GIS, gave each of 42,170 Manhattan lots a five-year redevelopment probability. It was trained only on 2015-2020 changes in NYC MapPLUTO and applied to 2020 parcel data; no permits or certificates were used to train it or as inputs. The forecast was made afterwards, so this is a retrospective test, not a live one. The comparison is the planners' soft-site rule (lots built to well under their zoning).

How ArcGIS Pro turned the records into a fair test. The same steps, with one dataset of new-building permits, give the second outcome. A ProveIt check (custom, not built into Pro) warned about the street-frontage problem in step 2.

Step in ArcGIS Pro	Tool	Result
1. Get the records	ProveIt, our custom Python toolbox for Pro: download with source notes and known traps	1,772 new-building certificates of occupancy, 2021-2025, in a box around Manhattan; about 600 in Manhattan
2. Plain overlay	Spatial Join (inside)	Only 30 certificates fell inside a lot: **16 completed lots**. Records are placed on the street frontage
3. Join by tax lot	Key join on the tax-lot number (BBL)	547 certificates matched 398 lots by their own number

Step in ArcGIS Pro	Tool	Result
4. Nearest lot for the rest	Generate Near Table, 30 m limit	54 certificates without a matching number, mostly condominium billing lots, go to the nearest lot: **430 candidate completed lots**. Check: 41 of the 54 land on their own tax block, 3 have no number, 10 on a neighbouring block; dropping those 10 changes every AUC by less than 0.01
5. Apply a building-age filter to reduce timing bias	Definition query: no building from 2015 or later in 2020	215 of the 430 lots had a recent building and were removed: **215 candidate completed lots remain** (210 once lots without a soft-site value are left out)

Results: the timing filter matters as much as the model. AUC is the chance that a lot that did redevelop is ranked above one that did not (0.5 is chance). Both methods are scored on the same lots: the 272 lots without a soft-site value are left out, which is why 150 permit lots become 146 and 215 certificate lots become 210. For certificates, the building-age filter applies to every lot: 500 lots with a building from 2015 or later are removed (215 of them certified), leaving 41,670, and 271 of those lack a soft-site value, leaving 41,399. The 95% confidence interval for the difference uses DeLong's method for two AUCs on the same lots. The rule ranks permits correctly, so its low score on certificates is not a reversed scale. The comparison is highly sensitive to the timing filter: the stricter filter moves the rule from 0.39 to 0.70 and the AI from 0.67 to 0.75.

Outcome, 2021-2025 (same lots for both)	AI forecast	Rule of thumb	Difference, 95% CI	Read as
New-building permits issued (146 of 41,898 lots)	AUC 0.82	AUC 0.81	-0.02 to 0.05	Similar; no clear difference
New-building certificates of occupancy (210 of 41,399 lots)	AUC 0.67	AUC 0.39	0.24 to 0.33	AI ranks completions better; the rule scores below chance
Same, stricter timing: lots with any building from 2000 or later removed (83 of 39,745 lots)	AUC 0.75	AUC 0.70	-0.01 to 0.10	AI higher, but the difference is uncertain

In plain numbers. Of the roughly 400 lots each method ranked highest (top 1%), the AI's list held 14 lots that received a new-building permit against about 11 for the rule, and 14 with a new-building certificate against about 7, under either timing filter. Picking lots at random would find 1 to 2. The counts are small and carry no interval.

Read honestly. The test checks where, not how many. Permits are not construction starts and certificates are not every completion; both are records with known gaps. The model also counts large enlargements (floor area up 25% and 2,000 sq ft) as redevelopment, while these records cover new buildings only, so the test covers the main part of what the model predicts. Few lots redevelop (about 0.4% received a new-building permit), so even well-ranked lots mostly did not change: the ranking can prioritize review, not predict single lots. The timing filter is a proxy, and agreement with past records is not cause and effect. The value shown is catching evaluation errors before a model's performance is reported, not a change in a city decision.

Fifteen-minute plan. One case, one workflow.

Minutes	Content
0-2	The planning question: which lots will redevelop? The retrospective design and the rule of thumb
2-6	Live in ArcGIS Pro: why a plain overlay finds 16 completed lots, and tax-lot plus nearest-lot matching finds 430 candidates
6-10	Timing: a building-age filter, and how much it changes the comparison
10-13	AI against the rule on the same lots; map where the forecast missed
13-15	The workflow and a one-page checklist; QR codes to the code and the land-use impact tool
15-20	Questions

Takeaways. A tested ArcGIS Pro workflow for evaluating parcel-level planning predictions against later public records, notes on 16 public outcome sources and their traps, and a one-page checklist. Code and results: github.com/ParyaPayami/proveit. Data: NYC MapPLUTO, DOB NOW job filings, DOB certificates of occupancy; PlanArena manuscript (Payami, 2026).